Analytical Cume
January 2011

(Note… equations at end of cume)

Separations and Liquid Chromatography (LC)

1. (50 points) Separation of a commercially-synthesized epoxy resin (polymer) on a size-exclusion column with a porous silica packing produced peaks in a chromatogram with the following retention times. The column is 6.2 mm diameter x 250 mm long, with a mobile phase of tetrahydrofuran and flow rate of 1.3 mL/min. The solid portion of the particles occupies 3.0% of the total volume of the column, the volume within the pores occupies 26.8%, and the volume between particles occupies 70.2%. Assume that this is an ideal exclusion column (there is no adsorption). The part of the resin oligomers that do not include the repeating unit (i.e. the end groups) add up to MW of 591. Polymers containing repeating units of 264 for n = 0, 1, 2, 3, etc. are possible. Below is a table of the results of a separation of a mixture of polymer oligomers. (Note: MW = molecular weight or molecular mass).

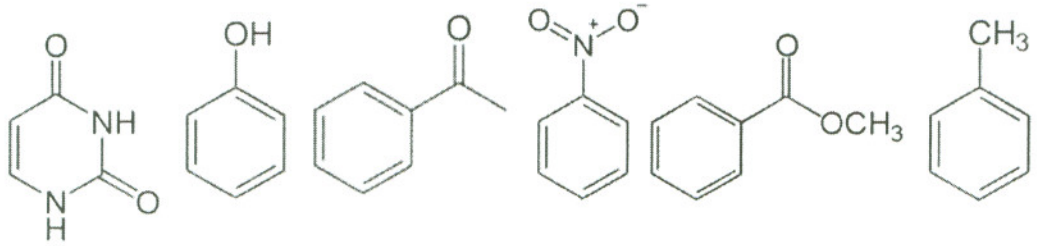
Species (MW = 591 + n x 264)	Retention Time / min
n=0	5.65
n=1	5.24
n=2	4.90
n=3	4.71
n=4	4.51
n=5	4.35
n=6	4.20

(a) (5 points) What is the total volume of the column in units of mL?
(b) (5 points) What is the volume in the pores in units of mL?
(c) (5 points) What is the volume around the outside and between the particles in units of mL?
(d) (10 points) Now that you know these volumes, at what retention volume would you expect molecules to elute that are too large to enter the pores (the exclusion limit) so that they elute together? Convert this retention volume into a retention time.
(e) (10 points) At what retention volume would you expect the molecules to elute that are so small, compared to the pore size, that they are no longer separated by the column (the permeation limit) and elute as a single band? Convert this retention volume into a retention time.
(f) (15 points) In general, what is the relationship of molecular weight (or molecular mass) to retention volume for a size exclusion column? At what time would you expect the n=7 oligomer to elute? Would you be able to resolve that peak? Why or why not?

2. (36 points) The following table provides a summary of the retention times of several chemical species eluted from partition chromatography of a mixture on a HPLC column containing 5-μm diameter particles. The length of the column is 20.0 cm, its diameter is 0.5 cm, the velocity of the mobile phase is 1.0 mL/min, volume of the stationary phase (V_s) is 1.79 mL, and the volume of the mobile phase (V_m) is 2.14 mL. The following data pertain to the separation (note W = width of peak at baseline) and t_R is retention time):

species	t_R, min	W, min
non-retained	0.8	---
uracil	2.5	0.10
phenol	4.6	0.20
acetophenone	6.4	0.50
nitrobenzene	8.0	0.60
methyl benzoate	11.6	0.70
toluene	22.5	1.10

Structures of the chemical species are shown below.



Uracil, phenol, acetophenone, nitrobenzene, methyl benzoate, toluene

(a) (6 points) What type of HPLC column, normal or reversed phase, was used to achieve the order of elution listed above? Why?

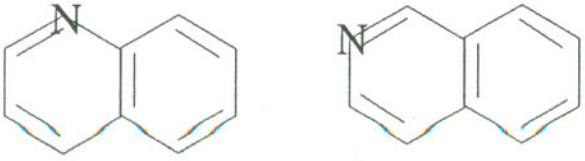
(b) (cont.) (30 points) Calculate the following:

the number of plates for uracil
the retention factor (or capacity factor) for uracil
the selectivity factor between uracil and phenol
the resolution for uracil and phenol. Are these peaks baseline resolved?
The length of the column to separate uracil and phenol so that they are just barely baseline resolved at $R_s = 1.5$.
What would be the time required to separate the two species on this shorter column?

3. (14 points) What type of LC would be best to separate the following, name the type of chromatography that you would use. Why?

(a) (7 points) an aqueous mixture of Tl^+, Cs^+, K^+, NH_4^+, Na^+, H^+, Li^+, Pb^{2+}, UO_2^{2+}, Br^-, NO_3^-, and OH^-.

(b) (7 points) water-insoluble isomers that are less than 2000 molecular weight and have the following general structures:



Equations that Might or Might Not be Helpful on Analytical Cume

$u = L/t_m$

$v = L/t_R$

$v = u \bullet$ (moles of solute in mobile phase/total moles solute)

$$\sigma^2 = H\,x$$

$$\sigma^2 = H\,L$$

$$N = \frac{L}{H}$$

$$N = \frac{L}{H} = \frac{L^2}{\sigma^2} = \frac{16\,L^2}{w^2}$$

$$N = \frac{16\,t_r{}^2}{w^2} = \frac{5.55\,t_r{}^2}{w_{1/2}{}^2}$$

$$H = A + \frac{B}{u} + C_{SP}\,u + C_{MP}\,u$$

$K = C_s/C_M$

k' = (time solute spends in the stationary phase / time solute spends in the mobile phase)

$$k'_A = \frac{\text{moles}_s}{\text{moles}_M} = \frac{C_s V_s}{C_M V_M} = K_A \frac{V_s}{V_M} = \frac{(t_{r,A} - t_M)}{t_M}$$

$$\alpha = \frac{t_{r,B} - t_m}{t_{r,A} - t_m} = \frac{t'_{r,B}}{t'_{r,A}} = \frac{k'_B}{k'_A} = \frac{K_B}{K_A} \quad \text{where } t'_{r,B} > t'_{r,A}$$

$$R_S = \frac{2\left(t_{r,B} - t_{r,A}\right)}{w_A + w_B} = \frac{\Delta t_r}{w_{av}}$$

$$R_S = \frac{\sqrt{N}}{4}\left(\frac{\alpha - 1}{\alpha}\right)\left(\frac{k'_b}{1 + k'_{av}}\right)$$

$$t_{r,B} = \frac{16\,R_S^2\,H}{u}\left(\frac{\alpha}{\alpha - 1}\right)^2\left(\frac{\left(1 + k'_b\right)^3}{\left(k'_b\right)^2}\right)$$